higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING SYSTEMS
NQF LEVEL 2

(6021022)

27 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Give ONE term for each of the following descriptions by choosing a term from the list below. Write only the term next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

lubrication; backlash; maintenance; clutch; swelling; cam

1.1.1 Small amount of free movement between the screw and nut

1.1.2 Disc with an irregular profile which drives a follower

1.1.3 Softening of a belt caused by fluids reaching the belt

1.1.4 Oiling machines with a slippery fluid to reduce friction between parts to make it last longer

1.1.5 Upkeep of equipment or machines

(5 × 1) (5)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A microprocessor is a small computer also called a chip or IC.

1.2.2 Torque is also called the turning moment.

1.2.3 Downtime is when a machine is working and producing products.

1.2.4 Stepper motors are DC motors that can be rotated in either direction.

1.2.5 Clean air filters lead to overheating and loss of operating pressure.

(5 × 1) (5)

1.3 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Rate at which work is done

1.3.2 Disconnecting a machine from its energy source to prevent it from being turned on accidentally

1.3.3 Item that allows electricity to flow through it

1.3.4 Output mechanism that changes hydraulic liquid pressure into mechanical movement

1.3.5 Restricts the flow of electricity in a circuit

(5 × 1) (5)

[15]

QUESTION 2

2.1 Draw each of the following electrical symbols:

2.1.1 Battery

2.1.2 Volt meter

2.1.3 Diode

2.1.4 Open switch

2.1.5 Capacitor

(5 × 1) (5)

2.2 Name the component or instrument used for each of the following:

2.2.1 Removing solids or particles from lubricating oil

2.2.2 Taking up any slackness in the chain when the distance is long

2.2.3 Storing compressed air until it is needed for the operation of the system

2.2.4 Stepping up or stepping down electrical voltage

2.2.5 Alerting, generating or cancelling signals in hydraulic systems

(5 × 1) (5)

2.3 Conductors and insulators are used in electrical systems.

Give TWO examples of each. (2 × 2) (4)

2.4 Give TWO uses of a transistor. (2)

2.5 Two resistors of 7 Ω and 8 Ω are connected in parallel and a 4 Ω resistor is then connected in series to the parallel circuit.

2.5.1 Draw a sketch of the circuit showing how the resistors are connected. (3)

2.5.2 Calculate the resistance of the parallel resistors. (4)

2.5.3 Calculate the total resistance of the circuit. (2)

[25]

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QUESTION 3

3.1 FIGURE 1 shows an electromagnetic brake.

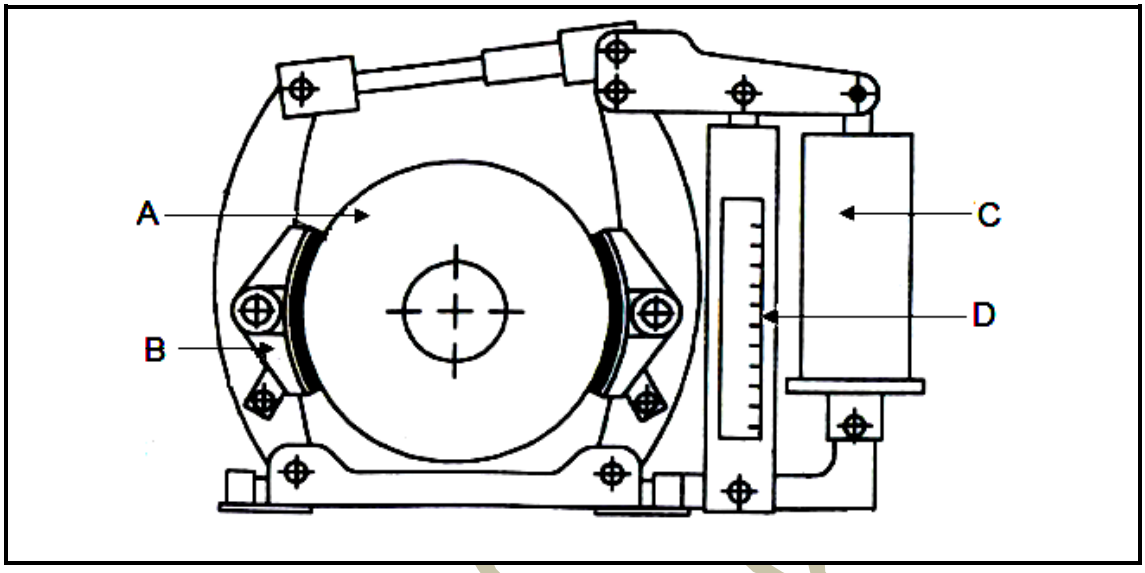


FIGURE 1

3.1.1 Label the parts of the electromagnetic brake by writing only the answer next to the letter (A–D) in the ANSWER BOOK. (4 × 1) (4)

3.1.2 Give the advantage of an electromagnetic brake in the event of a power failure. (2)

3.1.3 Give ONE application where electromagnetic brakes are used. (1)

3.2 Explain why belt drives cannot be lubricated. (4)

3.3 State the main differences between a *hydraulic* and a *pneumatic* system. (2)

3.4 Explain why gear drives are more efficient than belt drives. (4)

3.5 Name THREE main components that make up an air service unit in a pneumatic system. (3)

[20]

QUESTION 4

4.1 FIGURE 2 shows a compound gear drive with an input motor connected to gear W with 50 teeth rotating at 1 800 r/min. Gear X has 200 teeth, gear Y 40 teeth and gear Z 160 teeth.

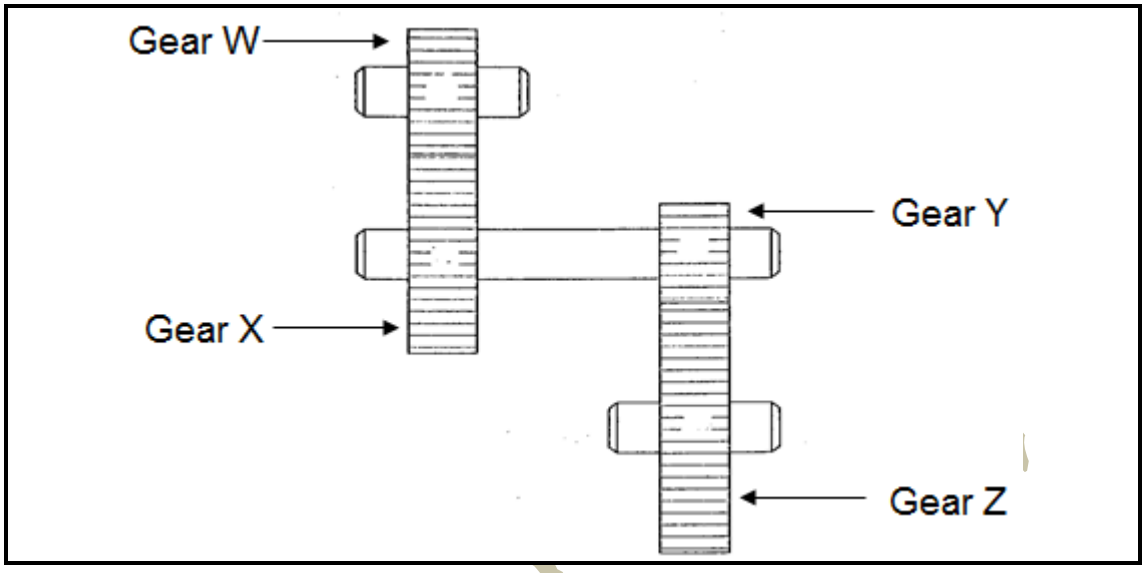


FIGURE 2

4.1.1 Name the driven gears. (2)

4.1.2 Name the input and output gears. (2)

4.1.3 Calculate the rotational speed of gear X. **Hint:** $N_A.T_A = N_B.T_B$ (2)

4.1.4 Calculate the rotational speed of gear Z. **Hint:** $N_A.T_A = N_B.T_B$ (2)

4.2 List FIVE basic hardware components found in a PLC system. (5)

4.3 List THREE types of braking systems used in machines. (3)

4.4 Show with the aid of a neat drawing the difference between a *spur* and a *helical* gear. (2 + 2) (4)

[20]

QUESTION 5

5.1 An electronic control system is made up of an input, a process and an output. The input is information supplied to the microprocessor by a sensor.

List THREE types of sensors used in engineering applications. (3)

5.2 List FIVE steps to take before returning a machine to operations after maintenance is completed. (5)

5.3 Belts must be inspected for wear and damage.

Name THREE defects to check for when examining belts for damage. (3)

5.4 Give THREE benefits of a good maintenance plan. (3)

5.5 List THREE functions of a control system. (3)

5.6 Give THREE reasons why it is important to record the tasks performed on equipment during maintenance on the job card. (3)

[20]

TOTAL: 100

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